

Practical Tools for Study Design and Inference

Chapter 4. Power Calculations and Sample Size Determination

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1. Background Ideas

Power of a Test

- Power is relevant to hypothesis testing

How likely is an alternative value or a difference to be detected?

- E.g., $H_0 : \mu = 1000$; $H_0 : \mu_F = \mu_M$; $H_0 : \beta_1 = 0$
- Common in medical and health studies. Setting desired power = 0.80 or 0.90 is typical.

Table 4.1: Terminology - Size and Power of a Test

Table 4.1: Terminology: size and power of a test

		Decision	
		Do not reject H_0	Reject H_0
State of nature	H_0 is true	Correct decision with probability $1 - \alpha$	Type I error—incorrect decision with probability α (<i>level</i> or <i>size</i> of test)
	H_0 is false (H_A is true)	Type II error—incorrect decision with probability β (at a specific alternative value)	Correct decision with probability $1 - \beta$ (<i>power</i> of test at a specific alternative value)

2. One-Sample Tests

Characterizing Hypotheses and Tests

- Hypotheses can be **simple** or **composite**.
- Tests can be characterized as **one-sided** or **two-sided**.
- When a hypothesis contains only one value, it is called simple (e.g., $H_0 : \mu = 3$)
- A hypothesis that contains more than one value is composite (e.g., $H_0 : \mu \leq 3$)
- Whether a test is one- or two-sided depends on the alternative. If the null hypothesis is $H_0 : \mu = 3$, a one-sided alternative is $H_A : \mu < 3$ or $H_A : \mu > 3$, because the alternative values of interest are only in one direction from the null value.

A two-sided alternative would be $H_A : \mu \neq 3$ since the alternatives can be either greater or less than $H_0 : \mu = 3$. The alternative, $H_A : \mu \neq 3$, is also composite, because it involves many values.

One-Sample Tests

- A single mean is tested against some hypothesized value(s) as

$$H_0 : \mu = \mu_0 \text{ versus } H_A : \mu = \mu_0 + \delta$$

at level α for some δ that can be positive or negative.

- Usually, we think of testing the simple H_0 vs the composite H_A :

$$H_0 : \mu = \mu_0 \text{ versus } H_A : \mu \neq \mu_0$$

- The standard test statistic:

$$t = \frac{\hat{y} - \mu_0}{\sqrt{v(\hat{y})}}, \text{ where } \hat{y} = \frac{\sum_{i \in s} w_i y_i}{\sum_{i \in s} w_i} \text{ and } w_i \text{ is the survey weight.}$$

- The t -statistic is referred to a normal distribution (or t). Validity depends on large sample approximation.
- The estimator of the variance of the mean needs to be consistent.

Figure 4.1: Size and Power of a One-sided Test

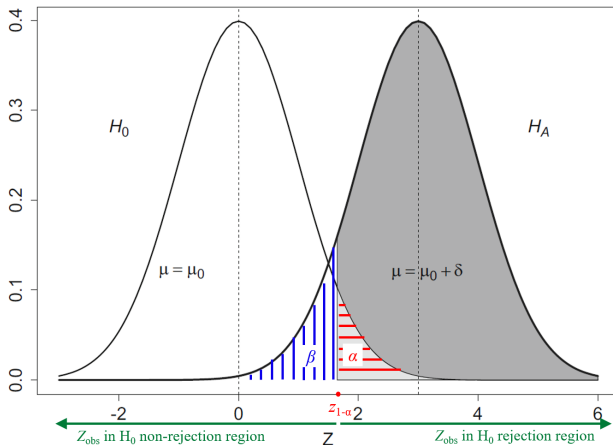


Fig. 4.1: Normal densities of test statistics under H_0 and H_A . $\delta / \sqrt{V(\hat{y})}$ is set equal to 3 in this illustration so that $E\{t | H_A \text{ is true}\} = 3$. A 1-sided test is conducted at the 0.05 level

Sample Size Calculations for Specified Power

- For a 1-sided test of $H_0 : \mu = \mu_0$ versus $H_A : \mu = \mu_0 + \delta$, the Type II error rate is $Pr(t \leq t_{1-\alpha}(df) | H_A \text{ is true}) = \beta$.
To actually compute β , we must think of a specific value within the possibilities spanned by H_A .
- **Power** is $1 - \beta$ (i.e., 1-Type II error rate):
 $Pr(t > t_{1-\alpha}(df) | H_A \text{ is true})$, the probability of correctly rejecting H_0 when it actually is false. The power and Type II error rate vary depending on the particular value of the alternative.
- For a 1-sided test of $H_0 : \mu = \mu_0$ versus $H_A : \mu > \mu_0$, the power is $Pr(t > t_{1-\alpha}(df) | \mu = \mu_0 + \delta) = 1 - \beta$.
- For a 2-sided test of $H_0 : \mu = \mu_0$ versus $H_A : \mu \neq \mu_0$, the power is $Pr(|t| > t_{1-\alpha/2}(df) | \mu = \mu_0 + \delta) = 1 - \beta$
i.e., the chance that the test statistic is in the rejectin region when $\mu = \mu_0 + \delta$.

Requirements for Computing Power for One Sample Tests

- The power for a given sample size depends on how far away the alternative value, $\mu_0 + \delta$, is from the null value, μ_0 . Four things are needed for a sample size calculation based on power:
 - ① Value of δ
 - ② Desired probability $1 - \beta$ (i.e., the power; $Pr(t > t_{1-\alpha}(df) | H_A : \mu = \mu_0 + \delta)$) of obtaining a significant test result when the true difference is δ ; and
 - ③ Significance level α of the test, which can be either 1-sided or 2-sided.
 - ④ Estimate of variance of the estimated mean. Translation may be needed to go from $v(\hat{y})$ to $(\sigma^2/n) \times \text{deff}$ so that n is exposed.

Sample Size Calculation - 1

- One-sided test

$$\begin{aligned}
 & Pr(t > t_{1-\alpha}(df) | \mu = \mu_0 + \delta) \\
 &= Pr\left(t - \frac{\delta}{\sqrt{V(\hat{y})}} > t_{1-\alpha}(df) - \frac{\delta}{\sqrt{V(\hat{y})}} | \mu = \mu_0 + \delta\right) \\
 &= Pr\left(\frac{\hat{y} - \mu_0}{\sqrt{v(\hat{y})}} - \frac{\delta}{\sqrt{V(\hat{y})}} > t_{1-\alpha}(df) - \frac{\delta}{\sqrt{V(\hat{y})}} | \mu = \mu_0 + \delta\right) \\
 &\doteq Pr\left(Z_{obs} > z_{1-\alpha} - \frac{\delta}{\sqrt{V(\hat{y})}}\right) = 1 - \beta \quad (4.4)
 \end{aligned}$$

Sample Size Calculation - 2

- By setting $z_{1-\alpha} - \frac{\delta}{\sqrt{V(\hat{y})}} = z_\beta$ with $V(\hat{y}) = \frac{\sigma_y^2}{n}$,

we solve for n :

$$n = \left[\sigma_y \frac{z_{1-\alpha} - z_\beta}{\delta} \right]^2 = \left[\sigma_y \frac{z_{1-\alpha} + z_{1-\beta}}{\delta} \right]^2 \quad (4.5)$$

through $z_{1-\alpha} - \frac{\delta\sqrt{n}}{\sigma_y} = z_\beta$

$$\frac{\delta\sqrt{n}}{\sigma_y} = z_{1-\alpha} - z_\beta$$

$$\text{and } z_\beta = -z_{1-\beta}$$

Sample Size Calculation - 3

- One-sided test

$$n = \left[\sigma_y \frac{z_{1-\alpha} - z_\beta}{\delta} \right]^2 = \left[\sigma_y \frac{z_{1-\alpha} + z_{1-\beta}}{\delta} \right]^2 \quad (4.5)$$

- Two-sided test

$$n = \left[\sigma_y \frac{z_{1-\alpha/2} - z_\beta}{\delta} \right]^2 = \left[\sigma_y \frac{z_{1-\alpha/2} + z_{1-\beta}}{\delta} \right]^2 \quad (4.9)$$

- $\frac{\delta}{\sigma}$ is also known as **Effect Size** (in one-sample test)
- What does a large effect size mean?

See **VDK Examples 4.1 and 4.4 in Class3Lab.pdf**

3. Differences in Means

3.1 Independent or Overlapping Samples

Differences in Means

- Compare means of two different groups of units. The term "two sample" test stems from the aim of comparing parameters for two separate groups or populations with a sample being selected from each.
- For a two-sample case (groups X and Y), we may want to test

$$H_0 : \mu_X \leq \mu_Y \text{ versus } H_A : \mu_X > \mu_Y$$

at level α .

- The sample test statistic is $t_d = \frac{\hat{d}}{\sqrt{v(\hat{d})}}$

with $\hat{d} = \hat{y} - \hat{x}$ and $v(\hat{d}) = v(\hat{x}) + v(\hat{y}) - 2\text{cov}(\hat{x}, \hat{y})$.

Differences in Means: Power

The null hypothesis that the mean of Y is larger than or equal to the mean of X ($H_0 : \mu_X \leq \mu_Y$) will be rejected in large samples if $t_d > z_{1-\alpha}$. If the true mean difference is $|\delta| > 0$, then the mean of t_d is $\frac{\delta}{\sqrt{V(\hat{d})}}$ instead of 0.

Letting $\mu_D = \mu_X - \mu_Y$, the probability of t_d in the rejection region is:

$$\begin{aligned} Pr\{t_d > z_{1-\alpha} | \mu_D = \delta\} &= Pr\left(t_d - \frac{\delta}{\sqrt{V(\hat{d})}} > z_{1-\alpha} - \frac{\delta}{\sqrt{V(\hat{d})}} \middle| \mu_D = \delta\right) \\ &\doteq Pr\left(Z > z_{1-\alpha} - \frac{\delta}{\sqrt{V(\hat{d})}}\right) \\ &= 1 - \beta \end{aligned} \quad (4.10)$$

Differences in Means: Variance

- Suppose sample size in each domain is the same and that the variance of the difference can be written as $V(\hat{d}) = \frac{\sigma_d^2}{n}$ where σ_d^2 is some population unit variance of the difference.
- With *srswor*, if the domain estimates are independent and their variances can be written as

$$V(\hat{d}) = \frac{\sigma_d^2}{n} = \frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{n} = \frac{1}{n}[\sigma_x^2 + \sigma_y^2].$$

- If the domain estimates are correlated, then $\sigma_d^2 = \sigma_x^2 + \sigma_y^2 - 2\sigma_{xy}$

$$V(\hat{d}) = \frac{\sigma_d^2}{n} = \frac{1}{n}[\sigma_x^2 + \sigma_y^2 - 2\sigma_{xy}].$$

- If $\sigma_x^2 = \sigma_y^2 \equiv \sigma_0^2$, then the unit-level correlation between y and x is $\rho = \sigma_{xy}/\sigma_0^2$ and $\sigma_d^2 = 2\sigma_0^2(1 - \rho)$ leading to

$$V(\hat{d}) = \frac{\sigma_d^2}{n} = \frac{2\sigma_0^2(1 - \rho)}{n}.$$

Sample Sizes for Testing Difference in Means

- To find the required sample size, we set z_β equal to $z_{1-\alpha} - \frac{\delta}{\sqrt{\sigma_d^2/n}}$ and solve for the sample size n to obtain

$$n = \left[\sigma_d \frac{(z_{1-\alpha} - z_\beta)}{\delta} \right]^2 \quad (4.11)$$

- Note that this is the sample size in **each** domain.
- If $\sigma_x^2 = \sigma_y^2 \equiv \sigma_0^2$ and $\rho = 0$ then $\sigma_d^2 = 2\sigma_0^2$.
- Compare (4.11) with (4.5)
- How about n for two-sided tests (i.e., $H_0 : \mu_X = \mu_Y$ versus $H_A : \mu_X \neq \mu_Y$)?
10pt]
- Use `power.t.test` with `type=c("two.sample")`

Two-Sided Test; Fig 4.4

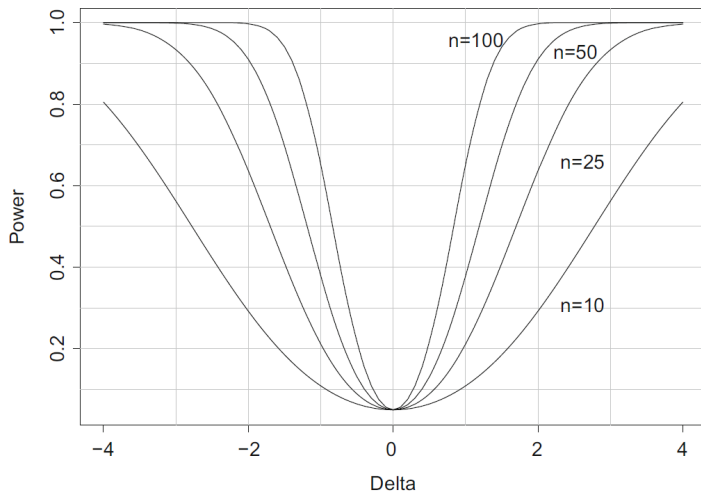


Fig. 4.4: Power for sample sizes of $n = 10, 25, 50, 100$ in a two-sided test of $H_0 : \mu_D = 0$ versus $H_A : |\mu_D| = \delta$ ($\alpha = 0.05$, $\sigma_d = 3$)

3.2 Partially Overlapping Samples

Partially Overlapping Samples

- Persons may be surveyed at baseline and then followed-up at a later time.
- Example: initial sample of n_1 is selected, a portion of these respond at time 2, and additional units are selected to obtain a total sample of n_2 for time 2.
- An estimate of the difference in population means may be desired, but the samples do not overlap completely because of dropouts, planned sample rotation, or nonresponse.
- Suppose s_1 and s_2 are sets of sample units with data collected only at times 1 and 2, and s_{12} denotes the overlap. Thus, the full samples at times 1 and 2 are $s_1 \cup s_{12}$ and $s_2 \cup s_{12}$.
- Assume that the samples at the two time periods are *srs*.
- Also assume that samples at times 1 and 2 are not necessarily the same size as $n_1 = rn_2$. The samples might be different sizes because of survey goal or data collection budget changes.

Difference and Its Variance

- Under *srs*, difference in means can be written as

$$\hat{d} = \hat{x} - \hat{y} = \frac{1}{n_1} \sum_{s_1} x_i - \frac{1}{n_2} \sum_{s_2} y_i + \sum_{s_{12}} \left(\frac{x_i}{n_1} - \frac{y_i}{n_2} \right)$$

- The variance can be expressed as

$$V(\hat{d}) = \frac{\sigma_x^2}{n_1} + \frac{\sigma_y^2}{n_2} - 2\sigma_{xy} \frac{n_{12}}{n_1 n_2} \quad (4.12)$$

With $n_{12} = \gamma n_1$ (i.e., γ : the proportion of overlap) and $r = n_1/n_2$,

$$V(\hat{d}) = \frac{1}{n_1} [\sigma_x^2 + r\sigma_y^2 - 2\gamma r\sigma_{xy}]$$

- For a 1-sided test of $H_0 : \mu_D = 0$ versus $H_A : \mu_D = \delta$, α level test, power = $1 - \beta$, the sample size n_1 is

$$n_1 = \frac{1}{\delta^2} [\sigma_x^2 + r\sigma_y^2 - 2\gamma r\rho\sigma_x\sigma_y] (z_{1-\alpha} - z_\beta)^2 \quad (4.13)$$

- No specific functions for partially overlapping sample tests in power analysis packages. Instead, use `nDep2sam` in `PracTools`.

See **VDK Example 4.7** in **Class3Lab.pdf**

4. Differences in Proportions

4.1 Independent Sample

Test Statistic - 1

- Test hypothesis $H_0 : P_1 = P_2$ where P_k is the population proportion in domain k .
- Assume independent *srs*'s are selected from each domain at n_1 and n_2 ; estimated proportions are p_1 and p_2 .
- If H_0 is true, $P_1 = P_2 = \bar{P}$. Variance of the difference is $V(p_1 - p_2) = \bar{P}(1 - \bar{P}) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)$

- Test statistic is then:

$$t_{\Delta p} = \frac{p_1 - p_2}{\sqrt{v(p_1 - p_2)}} \quad (4.16),$$

where $v(p_1 - p_2) = \bar{p}(1 - \bar{p}) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)$ and $\bar{p} = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}$ is a 'pooled' estimate of $\bar{P}$

- In large samples, $t_{\Delta p}$ is approximately normally distributed.

Test Statistic - 2

- Because variance of estimated proportions depends on their means, the arithmetic needed to get a power formula is a little different. Cover only the case of the same sample size n in each group (i.e., $n_1 = n_2 = n$).
- If the null hypothesis of equal proportions is true (i.e., $P_1 = P_2 = \bar{P}$), then

$$v(p_1 - p_2) = 2\bar{p}(1 - \bar{p})/n$$

- But, if $H_A : P_2 = P_1 + \delta$ is correct, $v(p_1 - p_2)$ does not depend on a pooled $\bar{p}$ but instead $v(p_1 - p_2) = (p_1q_1 + p_2q_2)/n$, where $q_1 = 1 - p_1$ and $q_2 = 1 - p_2$. This is an estimate of the theoretical variance,

$$V(p_1 - p_2) = (P_1Q_1 + P_2Q_2)/n$$

Power

- The power of this test for a 1-sided alternative $H_A : P_2 = P_1 + \delta$ is

$$\begin{aligned}
 & Pr(t_{\Delta p} > z_{1-\alpha} | P_2 - P_1 = \delta) \quad (\text{Note : } t_{\Delta p} = \frac{p_1 - p_2}{\sqrt{v(p_1 - p_2)}}) \\
 &= Pr \left(p_1 - p_2 > z_{1-\alpha} \sqrt{2\bar{P}(1 - \bar{P})/n} \mid P_2 - P_1 = \delta \right) \\
 &\doteq Pr \left(\frac{p_1 - p_2}{\sqrt{(P_1 Q_1 + P_2 Q_2)/n}} - \frac{\delta}{\sqrt{(P_1 Q_1 + P_2 Q_2)/n}} > \right. \\
 &\quad \left. \frac{z_{1-\alpha} \sqrt{2\bar{P}(1 - \bar{P})/n} - \delta}{\sqrt{(P_1 Q_1 + P_2 Q_2)/n}} \mid P_2 - P_1 = \delta \right) \\
 &= Pr \left(Z > \frac{z_{1-\alpha} \sqrt{2\bar{P}(1 - \bar{P})/n} - \delta}{\sqrt{(P_1 Q_1 + P_2 Q_2)/n}} \right) = 1 - \beta \quad (4.17)
 \end{aligned}$$

Sample Sizes

- The sample size in each group needed to detect a difference of δ is found by setting the right-hand side of the inequality in the last line of (4.17) equal to z_β and solving for n to give

$$n = \left[\frac{z_{1-\alpha}\Delta_1 - z_\beta\Delta_2}{\delta} \right]^2 \quad (4.18)$$

where $\Delta_1 = \sqrt{2\bar{P}(1 - \bar{P})}$ and $\Delta_2 = \sqrt{P_1Q_1 + P_2Q_2}$ (i.e., sqrt of the element variance under H_0 vs. H_A)

- Advance estimates of P_1 , P_2 , and $\bar{P}$ are needed to evaluate (4.18). The R function `power.prop.test` in `stats` package uses a search algorithm to solve for n which will give a similar answer to (4.18).

4.2 Overlapping Samples

Differences in Proportions: Overlapping Samples

- Suppose X and Y are equal to 1 with probabilities P_x and P_y
- The event that $Y = 1$ given that $X = 0$ could mean that a unit had a characteristic at time 2 given that it did not at time 1.
- Further suppose XY is equal to 1 with probability P_{xy} . The event, $XY = 1$, might correspond to a unit having some characteristic at both times 1 and 2. The conditional distribution of Y given X is $P_{y|x} = P_{xy}/P_x$; $P_{x|y}$ is defined similarly.
- With these definitions,

$$\sigma_x^2 = P_x(1 - P_x),$$

$$\sigma_y^2 = P_y(1 - P_y),$$

$$\sigma_{xy} = P_{xy} - P_x P_y, \text{ and}$$

$$\rho = \frac{P_{xy} - P_x P_y}{\sqrt{P_x(1 - P_x)P_y(1 - P_y)}}.$$

Differences in Proportions: Overlapping Samples

- When the sample sizes in the two groups are n_1 and n_2 , n_1 is found using (4.13)

$$n_1 = \frac{1}{\delta^2} [\sigma_x^2 + r\sigma_y^2 - 2\gamma r\rho\sigma_x\sigma_y] (z_{1-\alpha} - z_\beta)^2 \quad (4.13)$$

- Estimates (or educated guesses) are required for the proportions at the two time periods, P_x and P_y , and the proportion, P_{xy} , that retains the characteristic from the first time to the second.
- Use **nProp2sam** in **PracTools**. Two constraints on P_{xy} :
 - Since $P_{xy} = P_{y|x}P_x \leq P_x$ and $P_{xy} = P_{x|y}P_y \leq P_y$, it must be true that $P_{xy} \leq \min(P_x, P_y)$.
 - Since correlation must be in $[-1, 1]$, we must have

$$P_x P_y - \Delta \leq P_{xy} \leq P_x P_y + \Delta,$$

$$\text{where } \Delta = \sqrt{P_x(1 - P_x)P_y(1 - P_y)}$$

See **VDK Examples 4.8 and 4.10 in Class3Lab.pdf**

Special: Simulation-Based Power Analysis

Power for Complex Analysis

- Power is specific to a particular statistical test (e.g., t -tests of means, regression coefficients, etc.)
- For complex analysis (e.g., model coefficients), we may not have closed-form formulas for power analysis. Instead, simulate your data generating process and compute power as follows
 - ① Generate random data using the underlying model: specified design (e.g., sample/study design and sample size); values of covariates; parameter values that generate Y under the alternative hypothesis being tested; values of model random errors
 - ② Estimate the model (Or conduct the hypothesis tests)
 - ③ Check the p -values of the parameters of interest
 - ④ Repeat Steps 1-3 many times and observe the p -values
 - ⑤ Power=Proportion of p -value $< \alpha$ out of the repeats

See Example Simulation 1 and 2 in Class3Lab.pdf